

Commerce XI Maths App

Theory (Pages 45–49)

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 4
Bivariate Frequency Distribution and Chi Square Statistic
Total questions: 45

Question 1 - MCQ · PG-45 · TH-Q1

A frequency distribution involving one variable is called

- A. univariate frequency distribution
- B. bivariate frequency distribution
- C. contingency table
- D. conditional distribution

Correct answer: univariate frequency distribution

Explanation and steps

Univariate data involve one variable.

Question 2 - MCQ · PG-45 · TH-Q2

A frequency distribution of two variables observed on the same population is called

- A. bivariate frequency distribution
- B. univariate frequency distribution
- C. simple series
- D. time series

Correct answer: bivariate frequency distribution

Explanation and steps

Bivariate data record two variables together.

Question 3 - MCQ · PG-45 · TH-Q3

Which is an example of bivariate data?

- A. height and weight of individuals
- B. heights only
- C. weights only
- D. ages only

Correct answer: height and weight of individuals

Explanation and steps

Height and weight are observed simultaneously.

Question 4 - MCQ · PG-45 · TH-Q4

Income and expenditure of families form

- A. bivariate data
- B. univariate data
- C. categorical data only
- D. an ungrouped single variable

Correct answer: bivariate data

Explanation and steps

Two variables are recorded for every family.

Question 5 - MCQ · PG-45 · TH-Q5

A shopkeeper records daily temperature as X and the number of ice candies sold as Y.
Which variable represents daily temperature?

- A. X
- B. Y
- C. f_{ij}
- D. N

Correct answer: X

Explanation and steps

Daily temperature is represented by X.

Question 6 - MCQ · PG-45 · TH-Q6

A shopkeeper records daily temperature as X and the number of ice candies sold as Y.
Which variable represents the number of ice candies sold?

- A. Y
- B. X
- C. f_{ij}
- D. N

Correct answer: Y

Explanation and steps

The number of ice candies sold is represented by Y.

Question 7 - MCQ · PG-45 · TH-Q7

The 15 ordered pairs (X, Y), in day order, are:

(33, 250), (18, 150), (16, 250), (34, 400), (33, 500), (18, 250), (16, 150), (33, 500), (18, 150), (33, 500), (16, 250), (33, 400), (18, 400), (34, 500), (34, 400).

Which is the first observation?

- A. (33, 250)
- B. (16, 150)
- C. (18, 500)
- D. (34, 150)

Correct answer: (33, 250)

Explanation and steps

The first listed temperature is 33 and sales are 250.

Question 8 - MCQ · PG-45 · TH-Q8

The 15 ordered pairs (X, Y), in day order, are:

(33, 250), (18, 150), (16, 250), (34, 400), (33, 500), (18, 250), (16, 150), (33, 500), (18, 150), (33, 500), (16, 250), (33, 400), (18, 400), (34, 500), (34, 400).

At which position does the pair (16, 150) occur?

- A. seventh observation
- B. first observation
- C. second observation
- D. fifteenth observation

Correct answer: seventh observation

Explanation and steps

It is the seventh pair in the listed data.

Question 9 - MCQ · PG-45 · TH-Q9

A bivariate frequency table is also called a

- A. two-way table
- B. one-way table
- C. time chart
- D. stem plot

Correct answer: two-way table

Explanation and steps

The two variables define rows and columns.

Question 10 - MCQ · PG-45 · TH-Q10

Y X	16	18	33	34	Total
150	1	2	0	0	3
250	2	1	1	0	4
400	0	1	1	2	4
500	0	0	3	1	4
Total	3	4	5	3	15

Where are the values of X placed in this bivariate frequency table?

- A. horizontally at the top
- B. vertically on the left
- C. only in the total column
- D. outside the table

Correct answer: horizontally at the top

Explanation and steps

X forms the column headings.

Question 11 - MCQ · PG-45 · TH-Q11

Y X	16	18	33	34	Total
150	1	2	0	0	3
250	2	1	1	0	4
400	0	1	1	2	4
500	0	0	3	1	4
Total	3	4	5	3	15

Where are the values of Y placed in this bivariate frequency table?

- A. vertically on the left
- B. horizontally at the top
- C. only in the bottom row
- D. outside the table

Correct answer: vertically on the left

Explanation and steps

Y forms the row headings.

Question 12 - MCQ · PG-45 · TH-Q12

Each observed pair (x, y) contributes a tally to

- A. one corresponding cell
- B. every cell
- C. only the total cell
- D. the diagonal only

Correct answer: one corresponding cell

Explanation and steps

The intersection of its X and Y values identifies the cell.

Question 13 - MCQ · PG-46 · TH-Q13

Y \ X	16	18	33	34	Total
150	1	2	0	0	3
250	2	1	1	0	4
400	0	1	1	2	4
500	0	0	3	1	4
Total	3	4	5	3	15

The grand total is

- A. 15
- B. 8
- C. 12
- D. 20

Correct answer: 15

Explanation and steps

The data contain 15 daily observations.

Question 14 - MCQ · PG-46 · TH-Q14

Y \ X	16	18	33	34	Total
150	1	2	0	0	3
250	2	1	1	0	4
400	0	1	1	2	4
500	0	0	3	1	4
Total	3	4	5	3	15

The frequency corresponding to $X = 33$ and $Y = 500$ is

- A. 3
- B. 1
- C. 4
- D. 5

Correct answer: 3

Explanation and steps

The cell contains three observations.

Question 15 - MCQ · PG-46 · TH-Q15

Y \ X	16	18	33	34	Total
150	1	2	0	0	3
250	2	1	1	0	4
400	0	1	1	2	4
500	0	0	3	1	4
Total	3	4	5	3	15

The marginal frequency for $Y = 250$ is

- A. 4
- B. 1
- C. 3
- D. 5

Correct answer: 4

Explanation and steps

The row total for $Y = 250$ is 4.

Question 16 - MCQ · PG-46 · TH-Q16

Y \ X	16	18	33	34	Total
150	1	2	0	0	3
250	2	1	1	0	4
400	0	1	1	2	4
500	0	0	3	1	4
Total	3	4	5	3	15

The marginal frequency for $X = 33$ is

- A. 5
- B. 3
- C. 4
- D. 8

Correct answer: 5

Explanation and steps

The column total for $X = 33$ is 5.

Question 17 - MCQ · PG-46 · TH-Q17

Row totals together with values of Y give the

- A. marginal frequency distribution of Y
- B. conditional distribution of X
- C. marginal distribution of X
- D. grand total only

Correct answer: marginal frequency distribution of Y

Explanation and steps

Row totals summarize Y.

Question 18 - MCQ · PG-46 · TH-Q18

Column totals together with values of X give the

- A. marginal frequency distribution of X
- B. conditional distribution of Y
- C. marginal distribution of Y
- D. grand total only

Correct answer: marginal frequency distribution of X

Explanation and steps

Column totals summarize X.

Question 19 - MCQ · PG-46 · TH-Q19

The sum of all row totals equals

- A. the grand total N
- B. one
- C. the number of rows only
- D. the number of columns only

Correct answer: the grand total N

Explanation and steps

Both sets of marginal totals sum to N.

Question 20 - MCQ · PG-46 · TH-Q20

X Y	16	18	33	34	Total
150	1	2	0	0	3
250	2	1	1	0	4
400	0	1	1	2	4
500	0	0	3	1	4
Total	3	4	5	3	15

The conditional distribution of X when $Y = 250$ is

- A. 2, 1, 1, 0
- B. 1, 2, 0, 0
- C. 0, 1, 1, 2
- D. 3, 4, 5, 3

Correct answer: 2, 1, 1, 0

Explanation and steps

Read across the $Y = 250$ row.

Question 21 - MCQ · PG-46 · TH-Q21

$\begin{array}{c} X \\ Y \end{array}$	16	18	33	34	Total
150	1	2	0	0	3
250	2	1	1	0	4
400	0	1	1	2	4
500	0	0	3	1	4
Total	3	4	5	3	15

The conditional total when $Y = 250$ is

- A. 4
- B. 3
- C. 5
- D. 15

Correct answer: 4

Explanation and steps

The $Y = 250$ row total is 4.

Question 22 - MCQ · PG-46 · TH-Q22

$\begin{array}{c} X \\ Y \end{array}$	16	18	33	34	Total
150	1	2	0	0	3
250	2	1	1	0	4
400	0	1	1	2	4
500	0	0	3	1	4
Total	3	4	5	3	15

For $X = 33$ or 34 , the conditional frequencies of $Y = 150, 250, 400, 500$ are

- A. 0, 1, 3, 4
- B. 1, 2, 1, 4
- C. 3, 4, 4, 4
- D. 0, 1, 1, 3

Correct answer: 0, 1, 3, 4

Explanation and steps

Add the $X = 33$ and $X = 34$ columns.

Question 23 - MCQ · PG-47 · TH-Q23

If X has m values and Y has n values, a bivariate table contains

- A. $m \times n$ cells
- B. $m + n$ cells
- C. $m - n$ cells
- D. m/n cells

Correct answer: $m \times n$ cells

Explanation and steps

Each X value pairs with each Y value.

Question 24 - MCQ · PG-47 · TH-Q24

The i th value of X is denoted by

- A. x_i
- B. y_j
- C. f_{ij}
- D. N

Correct answer: x_i

Explanation and steps

This is the standard notation.

Question 25 - MCQ · PG-47 · TH-Q25

The j th value of Y is denoted by

- A. y_j
- B. x_i
- C. f_{ij}
- D. N

Correct answer: y_j

Explanation and steps

This is the standard notation.

Question 26 - MCQ · PG-47 · TH-Q26

The frequency associated with the pair (x_i, y_j) is denoted by

- A. f_{ij}
- B. x_i
- C. y_j
- D. N only

Correct answer: f_{ij}

Explanation and steps

Cell frequencies use f_{ij} .

Question 27 - MCQ · PG-47 · TH-Q27

The grand total of all cell frequencies is denoted by

- A. N
- B. m
- C. n
- D. f_{ij} only

Correct answer: N

Explanation and steps

N is total frequency.

Question 28 - MCQ · PG-47 · TH-Q28

The marginal frequency distribution of X is obtained from

- A. column totals
- B. row totals
- C. diagonal cells
- D. only zero cells

Correct answer: column totals

Explanation and steps

Each column total belongs to an X value.

Question 29 - MCQ · PG-47 · TH-Q29

The marginal frequency distribution of Y is obtained from

- A. row totals
- B. column totals
- C. diagonal cells
- D. only nonzero cells

Correct answer: row totals

Explanation and steps

Each row total belongs to a Y value.

Question 30 - MCQ · PG-47 · TH-Q30

A frequency distribution of one variable for a specified value of the other is

- A. conditional frequency distribution
- B. marginal distribution
- C. univariate raw data
- D. class interval

Correct answer: conditional frequency distribution

Explanation and steps

The other variable supplies the condition.

Question 31 - MCQ · PG-48 · TH-Q31

Y \ X	10	11	12	13	Total
21	1	0	2	1	4
22	1	3	1	0	5
23	2	0	1	1	4
24	1	2	2	2	7
Total	5	5	6	4	20

The number of students is

- A. 20
- B. 16
- C. 24
- D. 25

Correct answer: 20

Explanation and steps

The grand total is 20.

Question 32 - MCQ · PG-48 · TH-Q32

X Y	10	11	12	13	Total
21	1	0	2	1	4
22	1	3	1	0	5
23	2	0	1	1	4
24	1	2	2	2	7
Total	5	5	6	4	20

The marginal distribution of age X is

- A. 5, 5, 6, 4
- B. 4, 5, 4, 7
- C. 5, 4, 5, 6
- D. 20, 20, 20, 20

Correct answer: 5, 5, 6, 4

Explanation and steps

Read the column totals.

Question 33 - MCQ · PG-48 · TH-Q33

X Y	10	11	12	13	Total
21	1	0	2	1	4
22	1	3	1	0	5
23	2	0	1	1	4
24	1	2	2	2	7
Total	5	5	6	4	20

The marginal distribution of G.K. marks Y is

- A. 4, 5, 4, 7
- B. 5, 5, 6, 4
- C. 4, 4, 5, 7
- D. 20, 20, 20, 20

Correct answer: 4, 5, 4, 7

Explanation and steps

Read the row totals.

Question 34 - MCQ · PG-48 · TH-Q34

Y \ X	10	11	12	13	Total
21	1	0	2	1	4
22	1	3	1	0	5
23	2	0	1	1	4
24	1	2	2	2	7
Total	5	5	6	4	20

The conditional distribution of age when G.K. marks are 23 is

- A. 2, 0, 1, 1
- B. 1, 3, 1, 0
- C. 1, 0, 2, 1
- D. 5, 5, 6, 4

Correct answer: 2, 0, 1, 1

Explanation and steps

Read the Y = 23 row.

Question 35 - MCQ · PG-48 · TH-Q35

Y \ X	10	11	12	13	Total
21	1	0	2	1	4
22	1	3	1	0	5
23	2	0	1	1	4
24	1	2	2	2	7
Total	5	5	6	4	20

The conditional distribution of G.K. marks when age is 11 is

- A. 0, 3, 0, 2
- B. 1, 1, 2, 1
- C. 2, 0, 0, 3
- D. 4, 5, 4, 7

Correct answer: 0, 3, 0, 2

Explanation and steps

Read the X = 11 column.

Question 36 - MCQ · PG-48 · TH-Q36

Y X	135–145	145–155	155–165	Total
35–45	2	2	0	4
45–55	2	1	1	4
55–65	4	4	4	12
Total	8	7	5	20

The marginal frequency distribution of height is

- A. 8, 7, 5
- B. 4, 4, 12
- C. 7, 7, 6
- D. 20, 20, 20

Correct answer: 8, 7, 5

Explanation and steps

Read the column totals.

Question 37 - MCQ · PG-48 · TH-Q37

Y X	135–145	145–155	155–165	Total
35–45	2	2	0	4
45–55	2	1	1	4
55–65	4	4	4	12
Total	8	7	5	20

The marginal frequency distribution of weight is

- A. 4, 4, 12
- B. 8, 7, 5
- C. 4, 8, 8
- D. 20, 20, 20

Correct answer: 4, 4, 12

Explanation and steps

Read the row totals.

Question 38 - MCQ · PG-48 · TH-Q38

X Y	135–145	145–155	155–165	Total
35–45	2	2	0	4
45–55	2	1	1	4
55–65	4	4	4	12
Total	8	7	5	20

The conditional distribution of height when weight is at least 55 kg is

- A. 4, 4, 4
- B. 2, 2, 0
- C. 2, 1, 1
- D. 8, 7, 5

Correct answer: 4, 4, 4

Explanation and steps

Read the 55–65 row.

Question 39 - MCQ · PG-48 · TH-Q39

X Y	135–145	145–155	155–165	Total
35–45	2	2	0	4
45–55	2	1	1	4
55–65	4	4	4	12
Total	8	7	5	20

The conditional distribution of weight when height lies from 135 to 145 cm is

- A. 2, 2, 4
- B. 4, 4, 12
- C. 2, 1, 4
- D. 8, 7, 5

Correct answer: 2, 2, 4

Explanation and steps

Read the 135–145 column.

Question 40 - MCQ · PG-49 · TH-Q40

A zero cell frequency means

- A. that pair did not occur
- B. the marginal total is zero
- C. the grand total is zero
- D. the variables are identical

Correct answer: that pair did not occur

Explanation and steps

No observation belongs to that combination.

Question 41 - MCQ · PG-49 · TH-Q41

Marginal frequencies can be checked because their sum must equal

- A. N
- B. m
- C. n
- D. 1

Correct answer: N

Explanation and steps

All observations are counted once.

Question 42 - MCQ · PG-49 · TH-Q42

A conditional distribution generally uses

- A. one row or one column selected by a condition
- B. all marginal totals only
- C. only the grand total
- D. no cell frequencies

Correct answer: one row or one column selected by a condition

Explanation and steps

The condition restricts one variable.

Question 43 - MCQ · PG-49 · TH-Q43

Before tabulation, bivariate observations are written as

- A. ordered pairs (x, y)
- B. single values only
- C. percentages only
- D. class marks only

Correct answer: ordered pairs (x, y)

Explanation and steps

Each unit has two associated measurements.

Question 44 - MCQ · PG-49 · TH-Q44

The main purpose of classification is to

- A. summarize paired observations clearly
- B. change the observations
- C. remove all frequencies
- D. make N equal to 1

Correct answer: summarize paired observations clearly

Explanation and steps

Grouping reveals the joint distribution.

Question 45 - MCQ · PG-49 · TH-Q45

Bivariate tables help study

- A. how two variables occur together
- B. only one variable's average
- C. only chronological change
- D. only geometric figures

Correct answer: how two variables occur together

Explanation and steps

Joint frequencies show their combined pattern.